strategies became predictable, with time-to-live values attached to each signal type. Audit logs clearly showed where each piece of context originated, when it was last updated, and which system asserted it.

Normalization also lowered data exposure. Instead of each PDP retrieving its own set of attributes, the enterprise PIP controlled the signal scope, providing only the data relevant to a specific policy request. This enhanced confidentiality and removed redundant data storage across applications.

Normalization also enhanced signal transmission security. With a single enterprise PIP managing all outbound and inbound context, every signal followed the same pattern of encryption, signing, and verification. This uniformity made it easier to verify provenance, streamlined audits, and lowered the risk of configuration drift that could expose data or compromise integrity.

Normalization also accelerated the adoption of fine-grained authorization across ACME. Before the enterprise PIP, each team that wanted to implement policy-based access control had to create its own data connectors and caching logic. With the normalization layer in place, a shared library provided the same normalized context API to all departments. Any system could connect with the organization's authorization infrastructure simply by linking the library and requesting the context it needed. This lowered development effort, promoted consistent policy design, and helped fine-grained authorization spread rapidly throughout ACME.

### 12.6.3 Extending SSF inside the organization

As ACME refined its normalization layer, the team noticed similarities between its internal signaling issues and the aims of the Shared Signals Framework (SSF). SSF's event-driven approach—where systems subscribe to updates about changes instead of polling for them—presented an appealing pattern for sharing context across ACME's platforms.

By adopting SSF-like semantics internally, ACME could use a single library to manage both inbound and outbound updates consistently. HR could publish employment status changes, the device service could send posture updates, and PDPs could subscribe to the events they care about. Each